REDD+ Nesting Feasibility

Scoping potential of nested REDD+ markets with risk-allocated deforestation mapping

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Contents


```
##           GEOS           GDAL         proj.4 GDAL_with_GEOS     USE_PROJ_H
##       "3.13.1"       "3.11.4"        "9.6.2"         "true"         "true"
##           PROJ
##        "9.6.2"
```

Introduction

The following spatial covariates were imported as potential drivers of deforestation risk. Covariates were merged between demographic and geographic datasets surrounding the project area and national level datasets beyond the project area in order to enable jurisdictional analysis.

Import data

```r
library(geodata)
library(tmaptools)
library(tmap)
library(sf)
sf::sf_extSoftVersion()
```

AOI Jurisdiction

```
          GEOS           GDAL         proj.4 GDAL_with_GEOS     USE_PROJ_H
      "3.13.1"       "3.11.4"        "9.6.2"         "true"         "true"
          PROJ
       "9.6.2"
```

```r
knitr::opts_knit$set(root.dir = here::here())

# AOI Jurisdictional Bdry
aoi_country = geodata::gadm(country="ECU", level=0, path="./03_Spatial_Data/AOI") |> sf::st_as_sf()
aoi_states  = geodata::gadm(country="ECU", level=1, path="./03_Spatial_Data/AOI") |> sf::st_as_sf()
st_write(aoi_country, "./03_Spatial_Data/AOI/aoi_country.shp", delete_dsn = T)
```

```
Deleting source './03_Spatial_Data/AOI/aoi_country.shp' using driver 'ESRI Shapefile'
Writing layer 'aoi_country' to data source
  './03_Spatial_Data/AOI/aoi_country.shp' using driver 'ESRI Shapefile'
Writing 1 features with 2 fields and geometry type Multi Polygon.
```

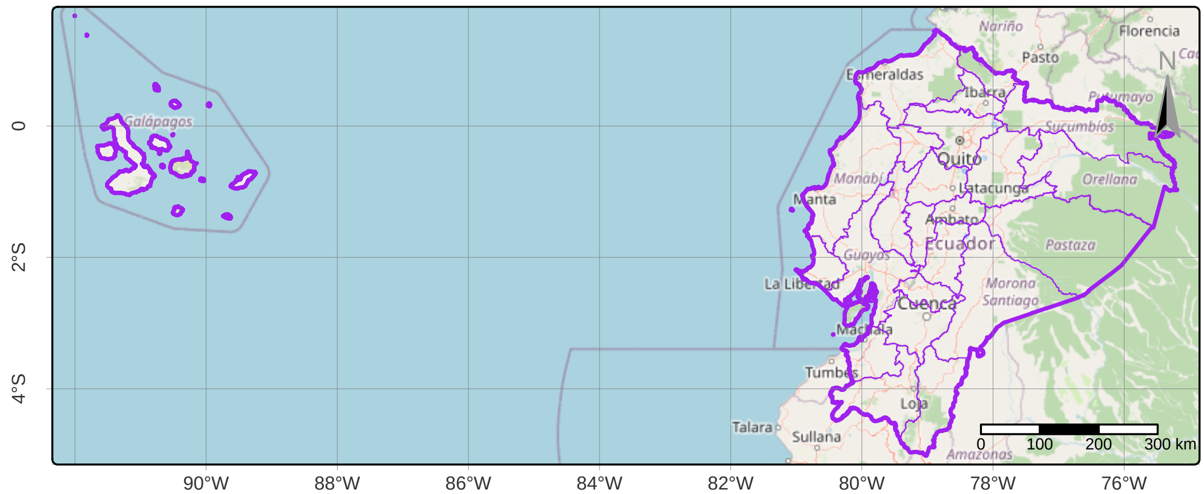
```r
st_write(aoi_states, "./03_Spatial_Data/AOI/aoi_states.shp", delete_dsn = T)
```

```
Deleting source './03_Spatial_Data/AOI/aoi_states.shp' using driver 'ESRI Shapefile'
Writing layer 'aoi_states' to data source
  './03_Spatial_Data/AOI/aoi_states.shp' using driver 'ESRI Shapefile'
Writing 24 features with 11 fields and geometry type Unknown (any).
```

```r
crs_master = sf::st_crs(aoi_country)

# Visualize
tmap::tmap_mode("plot")
tmap::tm_shape(aoi_country) + tmap::tm_borders(col="purple", lwd=2) +
  tmap::tm_shape(aoi_states) + tmap::tm_borders(col="purple", lwd=.5) +
  tmap::tm_scalebar(position=c("RIGHT", "BOTTOM"), text.size = .5) +
  tmap::tm_compass(color.dark="gray60",text.color="gray60",position=c("RIGHT", "top")) +
  tmap::tm_graticules(lines=T,labels.rot=c(0,90),lwd=0.2) +
  tmap::tm_layout(legend.position=c("left", "top"), legend.bg.color = "white") +
  tmap::tm_title("Risk Covariates Map: AOI Jurisdictional Boundaries", size=.8) +
  tmap::tm_basemap("OpenStreetMap")
```

Risk Covariates Map: AOI Jurisdictional Boundaries




```
#tmap::tm_basemap("OpenTopoMap")
#tmap::tm_basemap("Esri.WorldImagery")
```

```
library(osmdata)
library(httr2)
library(ggmap)

# Check OSM data & derive bbox window
osmdata::available_features()
osmdata::available_tags("place")
aoi_bbox = osmdata::getbb("Ecuador")

# Hospitals
buildings_hospital = aoi_bbox |> osmdata::opq() |>
  osmdata::add_osm_feature(key = "amenity", value = "hospital") |>
  osmdata::osmdata_sf() # |> st_transform(crs_master)

# Housing
buildings_residential = aoi_bbox |> osmdata::opq() |>
  osmdata::add_osm_feature(key = "building", value = "residential") |>
```

```r
  osmdata::osmdata_sf() # |> st_transform(crs_master)

# Religion
buildings_workship = aoi_bbox |> osmdata::opq() |>
  osmdata::add_osm_feature(key = "building", value = "religious") |>
  osmdata::osmdata_sf()

# Towns
townnames = aoi_bbox |> osmdata::opq() |>
  osmdata::add_osm_feature(key = "place", value = "town") |>
  osmdata::osmdata_sf()

# Villages
villages = aoi_bbox |> osmdata::opq() |>
  osmdata::add_osm_feature(key = "place", value = "village") |>
  osmdata::osmdata_sf()

# Merging collections
places_merged = townnames$osm_points |> bind_rows(villages$osm_points) |>
  group_by(across(-geometry)) |> sf::st_cast("POINT") |>
  summarise(geometry = st_union(geometry), .groups = "drop")
sf::st_write(places_merged, "./03_Spatial_Data/POP/places_merged.shp", delete_dsn=T)

buildings_merged = buildings_hospital$osm_points |>
  bind_rows(buildings_residential$osm_points, buildings_workship$osm_points) |>
  group_by(across(-geometry)) |> sf::st_cast("POINT") |>
  summarise(geometry = st_union(geometry), .groups = "drop")
sf::st_write(buildings_merged, "./03_Spatial_Data/POP/buildings_merged.shp", delete_dsn=T)

# Visualize
tmap::tm_shape(aoi_country) + tmap::tm_borders(col="purple", lwd=1) +
  tmap::tm_shape(aoi_states) + tmap::tm_borders(col="purple", lwd=.5) +
#  tmap::tm_shape(buildings_merged) + tmap::tm_symbols(size=0.15,lwd=0.5,fill="white",col="purple") +
#  tmap::tm_add_legend(type="symbols", col="white", fill="purple", size=0.8, labels="Buildings") +
  tmap::tm_shape(places_merged) + tmap::tm_symbols(size=0.2,lwd=0.5,fill="white",col="blue") +
  tmap::tm_add_legend(type="symbols", col="white", fill="blue", size=0.8, labels="Towns & Villages") +
  tmap::tm_scalebar(position=c("RIGHT", "BOTTOM"), text.size = .5) +
  tmap::tm_compass(color.dark="gray60",text.color="gray60",position=c("RIGHT", "top")) +
  tmap::tm_graticules(lines=T,labels.rot=c(0,90),lwd=0.2) +
  tmap::tm_layout(legend.position=c("left", "top"), legend.bg.color = "white") +
  tmap::tm_title("Risk Covariates Map: Built Environment", size=.8) +
  tmap::tm_basemap("OpenStreetMap")
  #tmap::tm_basemap("OpenTopoMap")
  #tmap::tm_basemap("Esri.WorldImagery")
```

Built Environment

```r
# Motorway/Highways
transport_motorway <- aoi_bbox %>% opq() %>%
```

```r
  add_osm_feature(key = "highway", value = "motorway") %>%
  osmdata_sf()

# Roads
transport_roads <- aoi_bbox %>% opq() %>%
  add_osm_feature(key = "XXXXXXX", value = "XXXXXXX") %>%
  osmdata_sf()

# Tracks
transport_tracks <- aoi_bbox %>% opq() %>%
  add_osm_feature(key = "XXXXXXX", value = "XXXXXXX") %>%
  osmdata_sf()

# Railways
transport_path <- aoi_bbox %>% opq() %>%
  add_osm_feature(key = "XXXXXXX", value = "XXXXXXX") %>%
  osmdata_sf()

# Merge all transport
transport_merged = transport_motorway |>
  bind_rows(transport_tracks, transport_path) |>
  group_by(across(-geometry)) |> sf::st_cast("MULTILINESTRING")
  summarise(geometry = st_union(geometry), .groups = "drop")
sf::st_write(transport_merged, "./03_Spatial_Data/Risk/transport_merged.shp", delete_dsn=T)

# Visualize
tmap::tm_shape(aoi_country) + tmap::tm_borders(col="green", lwd=3) +
  tmap::tm_shape(aoi_states) + tmap::tm_borders(col="orange") +
  tmap::tm_shape(buildings_merged) + tmap::tm_symbols(size=0.35,lwd=0.5,fill="",col="white") +
  tmap::tm_shape(transport_merged) + tmap::tm_symbols(size=0.35,lwd=0.5,fill="purple",col="white") +
  tmap::tm_add_legend(type="symbols", col="purple", fill="purple", size=0.8, labels="Buildings") +
  tmap::tm_scalebar(position=c("RIGHT", "BOTTOM"), text.size = .5) +
  tmap::tm_compass(color.dark="gray60",text.color="gray60",position=c("RIGHT", "top")) +
  tmap::tm_graticules(lines=T,labels.rot=c(0,90),lwd=0.2) +
  tmap::tm_layout(legend.position=c("left", "top"), legend.bg.color = "white") +
  tmap::tm_title("Risk Covariates Map: Built Environment", size=.8) +
  tmap::tm_basemap("OpenStreetMap")
  #tmap::tm_basemap("OpenTopoMap")
  #tmap::tm_basemap("Esri.WorldImagery")
```

Transportation

```r
waterways_country = sf::st_read("./03_Spatial_Data/HYDRO/waterways_country.shp")  #|>
  #sf::st_intersection(aoi_bbox) |>
  #dplyr::select(name,fclass) |>
  #dplyr::mutate(fclas=as.factor(fclass)) |>
  #dplyr::mutate(name=as.character(name))

```

```r
waterbodies_country = sf::st_read("./03_Spatial_Data/HYDRO/waterbodies_country.shp")
  sf::st_intersection(aoi_country) |>
  dplyr::select(name,fclass) |>
  dplyr::mutate(fclass=as.factor(fclass)) |>
  dplyr::mutate(name=as.character(name))

water_merged = waterbodies_country |>
  bind_rows(waterbodies_poly_lines, waterways_merged) |>
  group_by(across(-geometry)) |>
  summarise(geometry=st_union(geometry), .groups="drop")

water_merged = sf::st_cast(water_merged, "MULTILINESTRING")
sf::st_write(water_merged, "./03_Spatial_Data/HYDRO/water_merged.shp", delete_dsn=T)
```

Hydrography

```r
###################################
###################################
###################################
dem = raster::subset(STACK, "DEM")
###################################
###################################
###################################


slope_tangent = raster::terrain(dem,
  opt="slope",
  unit="tangent",
  neighbors=8,
  filename="./03_Spatial_Data/TOPO/slope_tangent.tif"
  )

slope_tangent = terra::rast("./03_Spatial_Data/TOPO/slope_tangent.tif")
slope_percent = slope_tangent * 100
slope_percent = terra::clamp(slope_percent, 0, 100)
slope_percent = raster::raster(slope_percent)
raster::writeRaster(slope_percent,"./03_Spatial_Data/TOPO/slope_percent.tif", overwrite=T)
```

Topography

Risk-allocated deforestation

Two methods were explored for weighting variables and creating a generalized deforestation risk index. For efficiency of time, we adopted the following risk indexing approach, which was based on a weighted sum of subjectively scored covariate effects. Please note this approach was also recommended for its ease of updating in subsequent iterations.

We applied this risk index to inform a risk weighted allocation of a 10-year deforestation rate, first by multiplying the fraction of pixel risk by zonal forest loss, and second by factoring out annual zonal loss by multiplying by pixel risk values, as shown below.[1]

$$\text{AllocatedLoss}_{\text{pixel}} = \left(\frac{\text{risk}_{\text{pixel}}}{\sum \text{risk}_{\text{zone}}} \right) \times \text{annual\_loss\_10yr}_{\text{zone}}$$

$$\text{allocated\_loss}_{\text{pixel}} = \text{risk}_{\text{pixel}} \times \left(\frac{\text{annual\_loss\_10yr}_{\text{zone}}}{\sum \text{risk}_{\text{zone}}} \right)$$

Both formulas describe the same operation in different orders of multiplication: each pixel in a given zone Z receives a share of `annual_loss_10yr` based on its proportional risk. This ensures that higher-risk pixels are allocated higher deforestation loss, which is in line with Verra guidance regarding allocated deforestation risk maps.

```r
#### assemble covariates
states        = sf::st_read("./03_Spatial_Data/RISK/aoi_states.shp")
districts     = sf::st_read("./03_Spatial_Data/RISK/aoi_districts.shp")
places        = sf::st_read("./03_Spatial_Data/RISK/places_merged.shp")
buildings     = sf::st_read("./03_Spatial_Data/RISK/buildings_merged.shp")
transport     = sf::st_read("./03_Spatial_Data/RISK/transport_merged.shp")
waterways     = sf::st_read("./03_Spatial_Data/RISK/water_merged.shp")
slope         = stars::read_stars("./03_Spatial_Data/RISK/slope_percent.tif")


tmap::tmap_mode("plot")
tmap::tmap_arrange(tm3, tm4, tm6, tm5, ncols=2)
```

Visualize Deforestation Risk

Visualize Deforestation Allocation

[1] We computed risk allocation based on each pixel's risk value relative to the sum of all pixel risks in that zone.